

The University of Azad Jammu and Kashmir,
Muzaffarabad



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Computer Networks

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Lab 02

Introduction to Computer Networks & Cisco Packet tracer

-Simple LAN-

Department Of Software Engineering

CREATING AND TEST SIMPLE LAN IN CISCO PACKET TRACER

PART-A

Network Topology Creation.

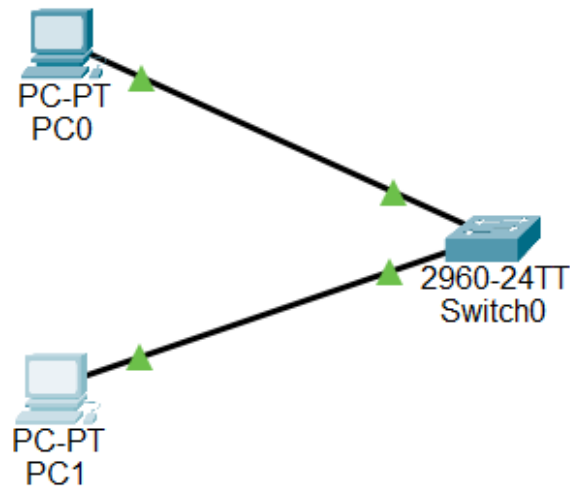


Figure 1

Description:

1. Two PCs are connected to Switch through wires.

PART-B

IP Address Configuration.

FOR PC0

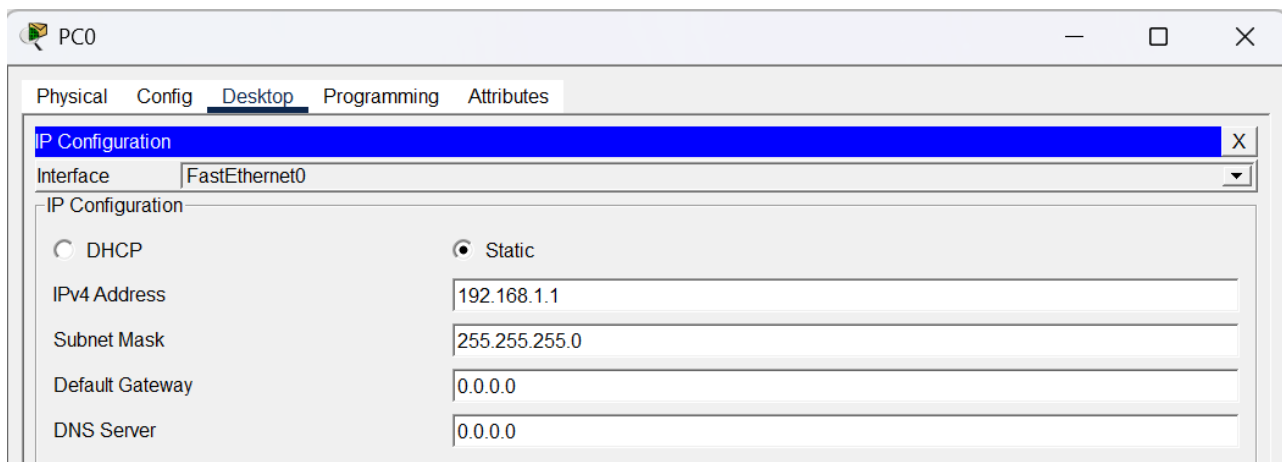


Figure 2: IP Configuration for PC0

FOR PC1

The screenshot shows a window titled "PC1" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Config" tab is active, and the "IP Configuration" window is open. The "Interface" dropdown is set to "FastEthernet0". Under "IP Configuration", the "Static" radio button is selected. The fields are filled with: IPv4 Address: 192.168.1.2, Subnet Mask: 255.255.255.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under "IPv6 Configuration", the "Static" radio button is also selected. The fields are filled with: IPv6 Address: (empty), Link Local Address: FE80::290:2BFF:FE31:8547, Default Gateway: (empty), and DNS Server: (empty).

Figure 3: IP Configuration for PC1

Description: 1. Configuration of IP address to both PCs.

PART-C Network Testing using CLI commands.

1.netstat Display active connection and statistics.

```
C:\>netstat

Active Connections

    Proto Local Address          Foreign Address         State
C:\>
```

Figure 4: Show active connections

2.ipconfig

Display the IP Configuration of PC.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::201:64FF:FED3:397D
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.1.1
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                                0.0.0.0
```

Figure 5: Display IP configuration

3.ping

Test connectivity between two devices.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=11ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms
```

Figure 6: Connectivity between two devices

4.tracert Show the path taken by packets.

```
C:\>tracert
Cisco Packet Tracer PC Tracert

Usage: tracert target

C:\>
```

Figure 7

5.arp -a Display ARP table (MAC address mapping).

```
C:\>arp -a
No ARP Entries Found

C:\>
```

Figure 8

Description: In this part, different Command Line Interface (CLI) networking commands were used to test and verify the network connectivity between the devices. Commands such as netstat, ipconfig, ping, tracert, and arp -a were executed to display network configuration details, check active connections, verify communication between PCs, observe the route taken by packets, and display the MAC address mapping in the ARP table.